447632

Course Name: **Discrete Mathematics**

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.

CO2: Analyze basic discrete structures and algorithms using effectively algebraic techniques.

CO3: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.

CO4: Understand the basic principles of sets, set equalities and operations in sets.

CO5: Apply counting principles to determine probabilities.

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Mid Term Examination, Odd Semester 2024-25
B.Tech(CS), II Year, III Semester
BCSC 1010 Discrete Mathematics

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	In a survey it was found that 21 people liked product A, 26 liked product B and 29 liked product C. If 14 people likes product A and B, 12 liked C and A , 14 people liked product B and C, and 8 liked all three products. a) Find how many people liked product C only. b) Find how many people like at least two product.	3	CO3	E	P
2	A drawer contains 12 red and 12 blue socks, all unmatched. A person takes socks out at random in the dark. How many socks must he take out to be sure that he has at least two blue socks?	3	CO5	An	C
3	Prove that for all $n \geq 1$, $7^n - 1$ is divisible by 6. In other words, prove that: is divisible by for all $7^n - 1$ is divisible by 6 for all $n \geq 1$	3	CO1	C	C
4	Suppose A is the set composed of all ordered pairs of positive integers. Let R be the relation defined on A where (a,b)R(c,d)means that a+d=b+c. Prove the relation is equivalence or not.	3	CO7	E	P
5	Course Enrollment In a university, students can enroll in courses. Define a function that maps each student to the courses they are enrolled in. Discuss whether this function can be both injective and surjective.	3	CO1	E	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
6	Solve the recurrence relation using generating functions: $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$ with initial conditions: $a_0 = 2$, $a_1 = 5$	5	CO2	E	P
7	Let A={1,2,3,4,6,24,36,72}, Let $\leq$ be the partial order defined by $A \leq B$ if a divide b. a) Draw hasse diagram. b) find the number of edges in hasse diagram. c) Prove hasse diagram is lattice or not. d) find join and meet of all element.	5	CO4	E	P
8	A tech company is organizing a conference and needs to arrange seating for 10 keynote speakers. These speakers consist of 4 data scientists, 3 AI researchers, and 3 cybersecurity experts. The company has specific rules for the seating arrangement: • All speakers must sit in a single row. • The 4 data scientists must sit together as a group. • The 3 AI researchers must also sit together as a group. • The 3 cybersecurity experts do not need to sit together but cannot be seated next to each other.	5	CO5	An	C